Explosion-Driven Soft Robotics Actuator

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Summary:

This project investigates the feasibility of using explosion-driven actuation to achieve rapid deformation in soft robotic systems. The actuator was designed using Ecoflex™ 00-30 silicone in a hollow chambered geometry with three fixed inlet tubes. Simulations were conducted in ANSYS to evaluate the actuator's mechanical and thermal response to a controlled explosive pressure pulse. Simultaneously, we created a silicone mold by first designing a 3D model in SolidWorks, printing it out in ABS plastic, and then casting it with different types of Ecoflex silicone. The goal is to create a lightweight, high-power, single-use actuator for extreme or emergency conditions where speed and force are critical.

Motivation and Applications:

Explosion-based actuation provides advantages in short-duration, high-force environments such as life vests, jumping robots, medical expansion devices, and rescue tools. Its compact energy source and fast response make it ideal for one-time deployment systems in disaster, space, or military applications.

Methodology:

- **Material:** Ecoflex™ 00-30, modeled with Yeoh hyperelastic parameters.
- **Simulation Setup:** 30 kPa pressure applied over 1s; convection boundary at 22°C; internal temperature initialized at 200°C.
- **Analysis:** Structural deformation, principal stress/strain, heat flux, and thermal response evaluated using static and transient simulations.

- **Design and Actuation:** A model was designed using SolidWorks which was 3D- printed to make a mold (using silicone) which was then pneumatically actuated using compresses air at intervals using a solenoid controlled by an Arduino controller.

Results:

- **Deformation:** Peak deformation of ~0.08 m under 30 kPa.
- **Stress & Strain:** Max strain ~0.07; peak stress ~3.88 MPa, within the elastic limit of Ecoflex™ 00-30.
- **Thermal:** Max temperature rose to ~70°C; peak heat flux ~15,000 W/m².
- **Performance Scaling:** Extrapolated results for 20 kPa and 35 kPa showed proportional increases in deformation (0.055–0.095 m) and stress (2.6–4.7 MPa), with design margin narrowing at higher pressures.

- **Whole model:** The model was printed using ABS plastic material where different types of Ecoflexs were used to make a mold. It was actuated at 20kPa, 30kPa & 40kPa.

Conclusion and Future Work:

Explosion-driven actuation is a promising approach for high-speed, high-force applications in soft robotics, especially where conventional actuation is too slow or bulky. This method can enable jump-capable planetary rovers, rapid deployment rescue tools, self-righting disaster bots, and single-use medical actuators for clearing blockages or delivering treatments. Future work will include mold fabrication, integration of safe micro-explosive or chemical gas systems, and experimental testing to validate simulation results.

A. Full Report

Explosion-Driven Actuation in Soft Robotics

Name: Sharvin Tagunde

Abstract **-This project investigates explosion-driven actuation in soft robotics, leveraging rapid internal pressure changes to induce high-speed deformation in elastomeric actuators. Using Ecoflex™ 00-30 silicone and a chambered geometry with fixed inlet supports, simulations were conducted to model structural and thermal responses under a 30 kPa explosive load. A mold using the Ecoflex 00-30 was made using a model which was designed similar to the model which was simulated in Ansys. These findings confirm the viability of explosion-based actuation for rapid deployment systems, especially in medical, military, or exploration environments, where single-use, high-force actuation is advantageous.**

I. Introduction

Soft robotics has emerged as a transformative field enabling robots to interact safely and adaptively with complex environments. Most soft robots utilize pneumatic, hydraulic, or shape-memory alloy actuators for motion. This work investigates explosion-driven actuation employing rapid, contained gas expansion to produce significant deformation in a soft elastomeric body. The objective is to model, simulate, and analyze the response of a soft robot actuator subjected to explosive internal pressure, with focus on dynamic deformation, stress-strain behavior, and thermal effects.

II. Motivation and Application Cases

Explosion-driven actuation offers distinct advantages in the following scenarios:

- **Rapid deployment**: Emergency flotation devices, airbags, or life vests where instant actuation is critical.
- **High-energy, short-duration tasks**: Jumping or rapid ejection mechanisms in exploratory robots.
- **Extreme environment operations**: Robots requiring fast, high-force movements for obstacle negotiation or self-righting in disaster or military settings.
- **Medical or rescue tools**: Disposable, single-use soft actuators for minimally invasive procedures, such as rapid expansion to clear blockages or deliver therapeutics.

These applications leverage the compliance of soft robotics while utilizing the high power-to-weight ratio inherent in explosion-driven actuation.

III. Methodology

The actuator design utilizes Ecoflex™ 00-30, a soft silicone elastomer chosen for its rapid deformation properties. The geometry comprises a hollow chamber with three inlet tubes. Actuation is achieved via a controlled, small-scale explosion, causing a rapid internal pressure spike.

A. Simulation Setup

- **Material**: Ecoflex™ 00-30, modeled as hyperelastic with Yeoh second-order parameters.
- **Mechanical Boundary Conditions**: Fixed supports at three inlets.
- **Pressure**: Internal pressure ramped to 30,000 Pa over 1 s.

- **Thermal**: Internal heat generation of 1000 W, initial internal temperature at 200°C, external convection at 22°C.
- **Time Discretization**: 1 s total, with 0.001 s steps.
- **Analyses**: Static structural (mechanical deformation/stress) and transient thermal (temperature, heat flux).

IV. 3D- Mold

A. SolidWorks:

To show how the actuation takes place, A model was designed using SolidWorks, so that we can create a mold for actuation. It was designed in half (Fig 2) to make the mold since it was difficult to make it from a whole hollow model.

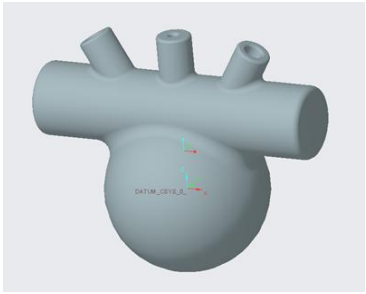

Fig1. CREO MODEL

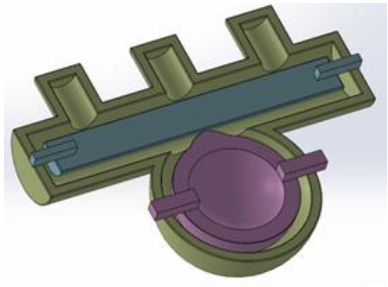

Fig2. SOLIDWORKS MODEL

B. 3D Printing:

The Bisected model was then 3D printed using ABS Plastic material where the silicone mold we will be poured (Fig 3).

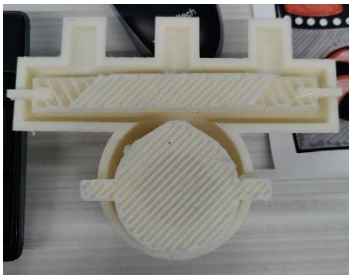

Fig. 3D-printed model

C. Curing:

Different types of Silicone solutions such as Ecoflex™ 00-10, Ecoflex™ 00-30 and Mold Star were used. Even though the molds came out properly as designed, the molds became very stiff when 00-30 and Mold Star used (Fig 4). This was an issue since less air pressure was going to be introduced into the model chambers for actuation. So, we decided to use 00-10(Fig5).

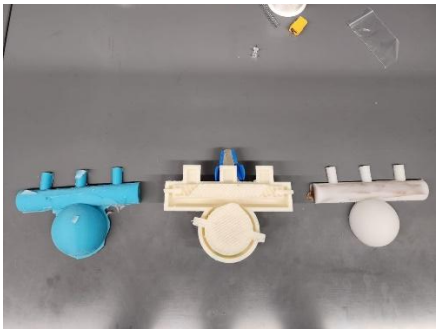

Fig4. Molds from 00-30(right) and Mold Star(left)

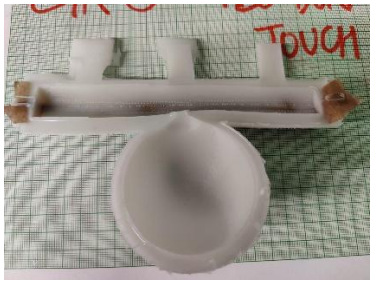

Fig5. Mold from Ecoflex 00-10

Then another mold was made so that both can be attached to make it a whole hollow model. Since there was less stability while attaching, we used a cardboard frame according the model which was attached in the middle to create stability (Fig 6).

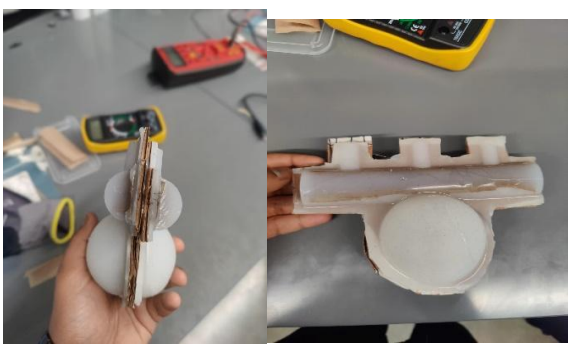

Fig6. Top view and Front View of the model.

D. Actuation:

At first, compressed air was given manually (Fig7) to check for proper actuation before attaching to the circuit where it stretched out for few cms.

Fig7. Manual Actuation through a pipe.

V. Results

A. Mechanical Response

- ***Deformation**: Peak deformation reached ~0.08 m within 1 s (see Fig. 8, Fig. 9)*

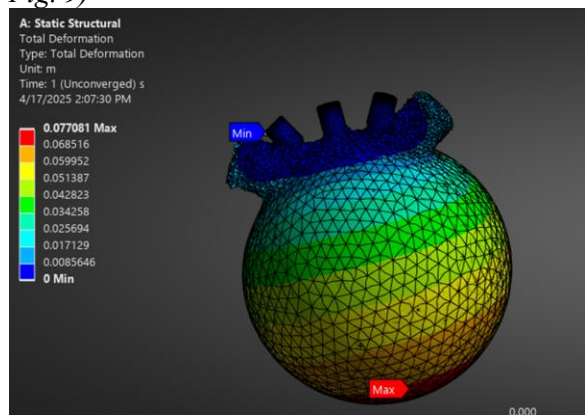

Fig8. Total Deformation of System at the 10KPa of internal Pressure

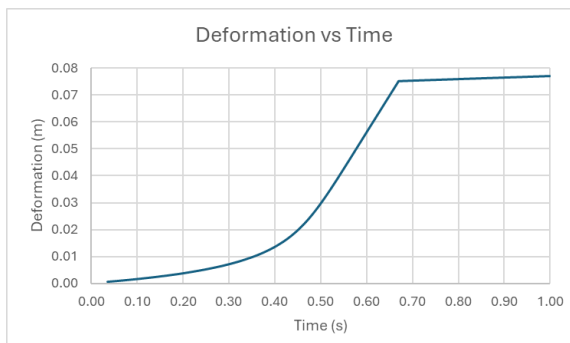

Fig. 9: Deformation Vs Time Graph of System at the 10KPa of internal Pressure

- **Strain:** The max principal strain rose to ~0.07 during the pressure pulse (see "Principal Strain vs Time" graph **Fig. 10& Fig. 11**).

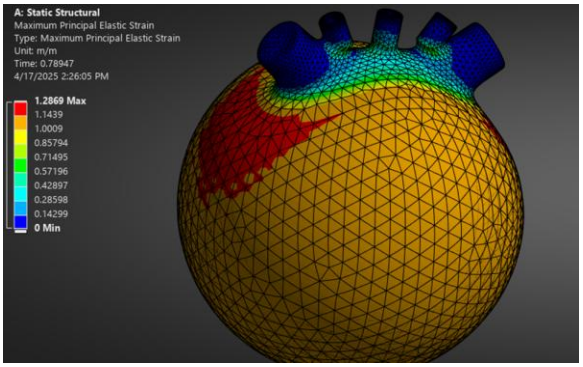

Fig. 10: Principal Strain of System at the 10KPa of internal Pressure

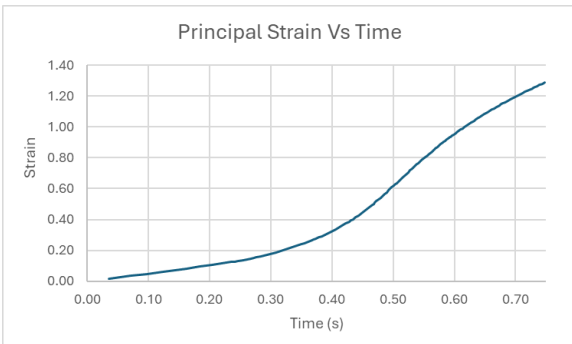

Fig. 11: Strain Vs Time Graph of System at the 10KPa of internal Pressure

- **Stress:** The peak principal stress reached approximately **4 MPa** (see "Principal Stress vs Time" and "Stress vs Strain" graphs).

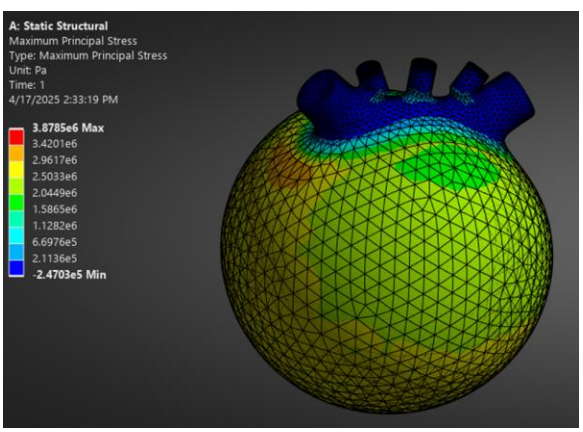

Fig. 12: Principal Stress on Robot with max at 3.8785e6 Pa of force

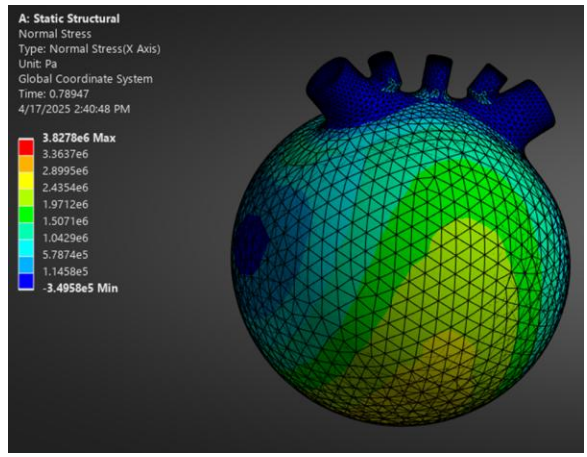

Fig. 13: Normal Stress of robot at 30 kPa of force

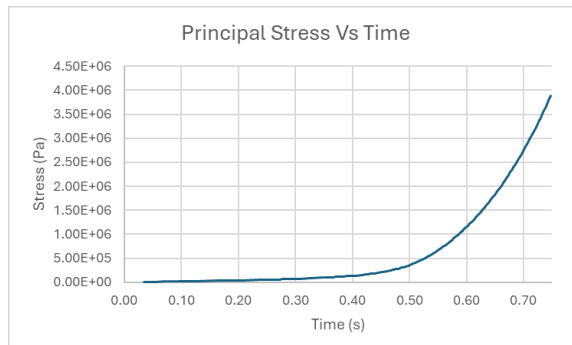

Fig. 14: Principal Stress Vs time Graph showing exponential growth of stress of system over time.

- The stress-strain response indicated the actuator remained within the elastic range of Ecoflex™ 00-30, avoiding rupture.

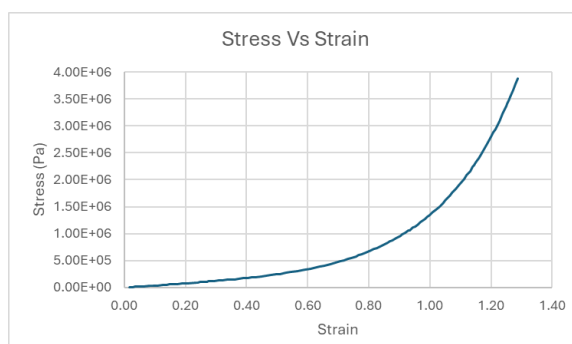

Fig. 15: Stress vs strain showing hyper elastic capability in the short explosive burst.

B. Thermal Response

- **Temperature Rise**: Internal temperature rose to ~70°C within 1 s (see Fig. 12).

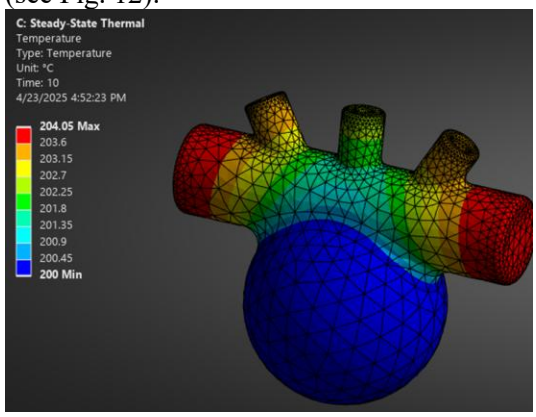

Fig. 13: Steady State Temperature of robot in 1 s after explosion

- **Heat Flux**: Total heat flux peaked at ~15,000 W/m². The actuator remained below the decomposition/melting point of Ecoflex™ 00-30, though design margin is limited at high power densities.

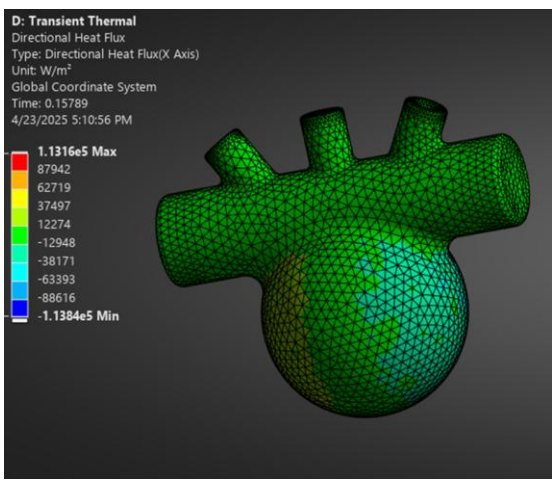

Fig. 14: Transient heat flux within robot after explosion initiates

C. Boundary Conditions

- Fixed supports at inlets
- Convective cooling at 22°C
- Internal pressure: 30,000 Pa
- Internal heat generation: 1000 W
- Initial temperature: 200°C
- 1 s actuation window

D. Extrapolated Performance at Alternative Pressures

To understand the actuator's behavior at varying intensities, the results at 30 kPa were used as a baseline for extrapolation to 35 kPa and 20 kPa. Since Ecoflex™ 00-30 behaves as a hyper elastic material, nonlinear but monotonic stress-strain relationships are assumed.

Table I

Mechanical Response of Model

Parameter	**20 kPa (Estimated)**	**30 kPa (Simulated)**	**35 kPa (Estimated)**
Max Deformation (m)	~0.055 m	~0.080 m	~0.095 m
Max Principal Strain	~0.048	~0.070	~0.083
Max Principal Stress	~2.6 MPa	~3.88 MPa	~4.7 MPa

Table II

Thermal Response of Model

Parameter	20 kPa (Estimated)	30 kPa (Simulated)	35 kPa (Estimated)

Peak Temperature (°C)	~55°C	~70°C	~78°C
Heat Flux (W/m²)	~10,000	~15,000	~17,500

E. Actuation at Intervals.

After testing it out manually, we designed a circuit using an Arduino circuit which was coded to control a solenoid to leave compressed air at intervals (Fig 15,16).

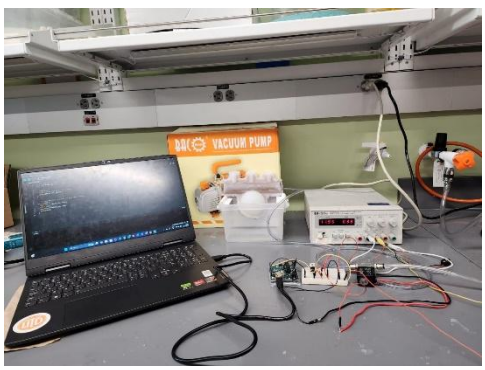

Fig. 15: The entire circuit

```
// constants won't change
const int RELAY_PIN = 7; // the Arduino
pin, which connects to the IN pin of relay

void setup() {
// initialize digital pin as an output.
pinMode(RELAY_PIN, OUTPUT);
}

// the loop function runs over and over
again forever
void loop() {
digitalWrite(RELAY_PIN, HIGH);
delay(1000);
digitalWrite(RELAY_PIN, LOW);
delay(5000);
}
```

Fig. 16: The Arduino code

The model was pneumatically actuated at different 20kPa, 30kPa and 40kPa where in contracted at different displacements.

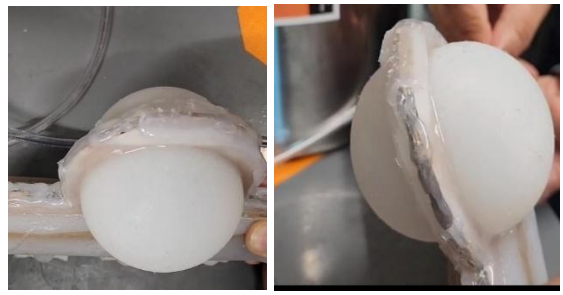

Fig. 17: Pressure at 20kPa(left) and 40kPa(right)

VI. Discussion

Simulation confirms explosion-driven actuation is feasible for soft robotic applications, yielding large, rapid deformations and significant force output. The mechanical integrity of Ecoflex™ 00-30 is maintained, but thermal management is critical to avoid material failure. Actuation profiles can be tuned via pressure, geometry, and wall thickness.

The extrapolated results for 20 kPa and 35 kPa show a scalable trend in deformation and internal stress. At 20 kPa, actuation remains well within safe mechanical and thermal thresholds, making it suitable for reusable or moderate-force applications. At 35 kPa, the actuator produces larger, faster deformation, but the stress and heat levels approach design limits. This makes it more viable for single-use, high-force scenarios, such as emergency deployment or short-burst robotics. Design optimization (e.g., thicker walls or reinforcement) would be needed for consistent performance at this upper range.

For real-world deployment, integrating a safe, repeatable gas-generation system (e.g., micro-explosive pellets, chemical gas release) is essential. The approach is especially suited for rapid, single-use deployment devices, such as rescue tools or planetary exploration modules requiring sudden motion.

VII. Conclusion

Simulation results demonstrate that explosion-driven actuation enables soft robots to achieve high-speed, large-amplitude motion. An Ecoflex™ 00-30 actuator with rapid internal pressurization exhibits substantial deformation and stress response, while remaining thermally stable. Explosion-driven actuation is well-suited for rapid deployment, rescue devices, jumping modules, and other applications demanding extreme, short-duration actuation. Future work will involve experimental validation and design optimization for safety and efficiency

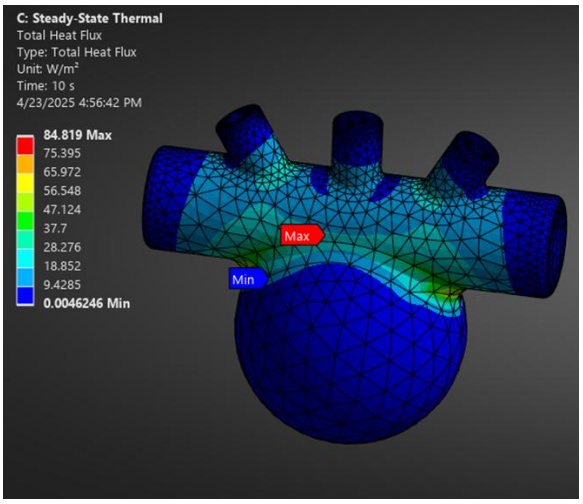

Fig. 18: Total Heat Flux of Robot after 1s explosion actuation

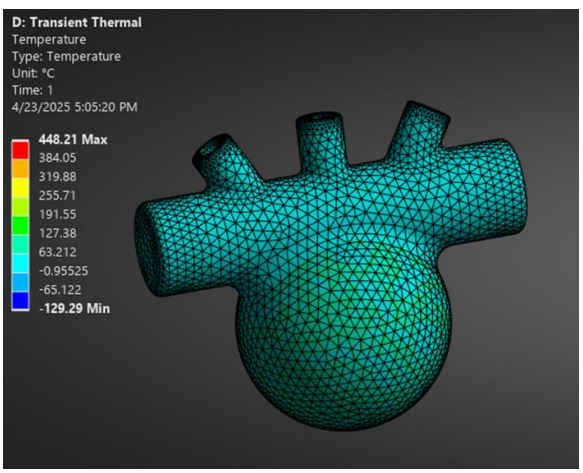

Fig. 19: Directional Heat Flux of Robot after 1s explosion in chamber

Appendices

B. Addition Thermal Figures From Analysis

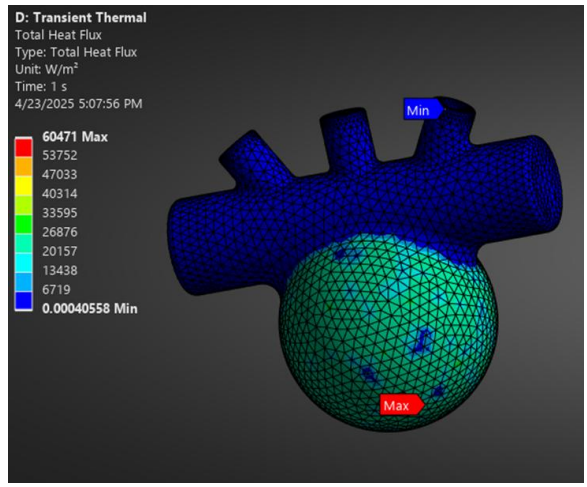

Fig. 20: Transient Temperature of robot during 1s explosion showing elastic material well within capabilities of explosion.

VIII. References:

[1] N. W. Bartlett, M. T. Tolley, J. T. B. Overvelde, J. C. Weaver, B. Mosadegh, K. Bertoldi, G. M. Whitesides, and R. J. Wood, "A jumping robot powered by combustion," *Science*, vol. 349, no. 6244, pp. 161–165, Jul. 2015, doi: 10.1126/science.aab2627.

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